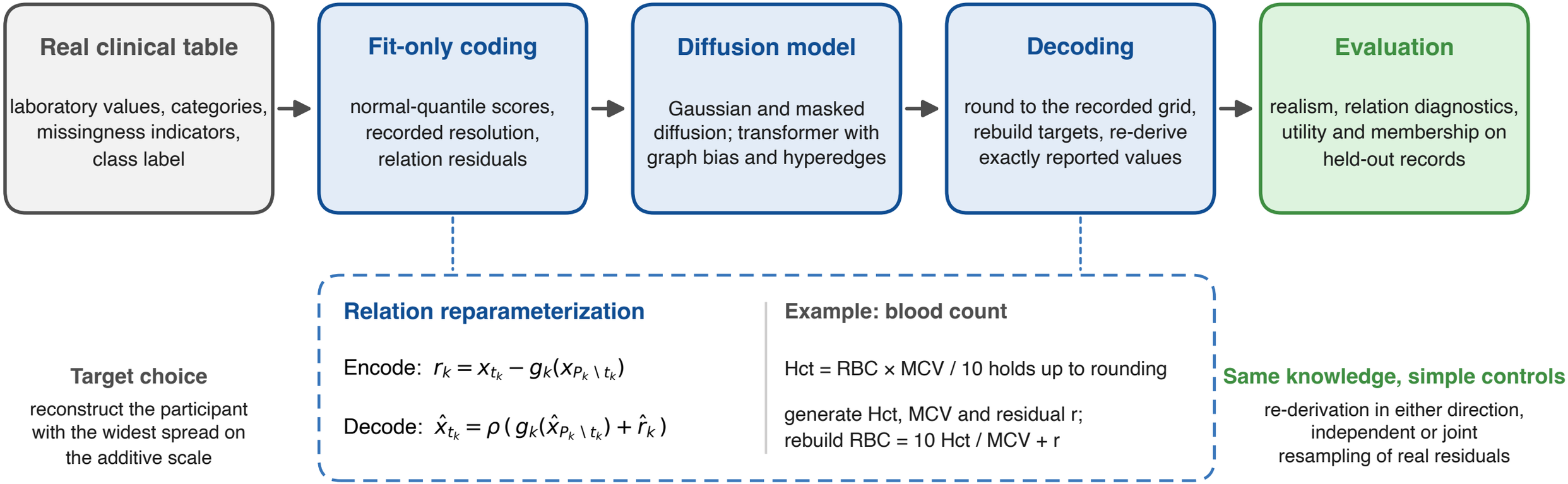
Real clinical table
laboratory values, categories, missingness indicators, class label
Fit-only coding
normal-quantile scores, recorded resolution, relation residuals
Diffusion model
Gaussian and masked diffusion; transformer with graph bias and hyperedges
Decoding
round to the recorded grid, rebuild targets, re-derive exactly reported values
Evaluation
realism, relation diagnostics, utility and membership on held-out records
Target choice
reconstruct the participant with the widest spread on the additive scale
Relation reparameterization
Encode: $r_k = x_{t_k} - g_k(x_{P_k \setminus t_k})$
Decode: $\hat{x}_{t_k} = \rho\,(\,g_k(\hat{x}_{P_k \setminus t_k}) + \hat{r}_k\,)$
Example: blood count
Hct = RBC × MCV / 10 holds up to rounding
generate Hct, MCV and residual r; rebuild RBC = 10 Hct / MCV + r
Same knowledge, simple controls
re-derivation in either direction, independent or joint resampling of real residuals